

Module-3

Eigen value & Eigen vector for Symmetric matrix

★ Jacobi Method:

$$A = \begin{bmatrix} a_{ii} & a_{ik} \\ a_{ki} & a_{kk} \end{bmatrix} \quad S = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$S^T A S = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} a_{ii} & a_{ik} \\ a_{ki} & a_{kk} \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos \theta a_{ii} + \sin \theta a_{ki} & \cos \theta a_{ik} + \sin \theta a_{kk} \\ -\sin \theta a_{ii} + \cos \theta a_{ki} & -\sin \theta a_{ik} + \cos \theta a_{kk} \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2 \theta a_{ii} + \sin \theta \cos \theta a_{ki} + \cos \theta \sin \theta a_{ik} + \sin^2 \theta a_{kk} & -\cos \theta \sin \theta a_{ii} - \sin^2 \theta a_{ki} + \cos \theta a_{ik} + \sin \theta \cos \theta a_{kk} \\ -\sin \theta \cos \theta a_{ii} + \cos^2 \theta a_{ki} - \sin \theta \cos \theta a_{ik} + \sin^2 \theta a_{kk} & \sin^2 \theta a_{ii} + \sin \theta \cos \theta a_{ki} - \sin \theta \cos \theta a_{ik} + \cos^2 \theta a_{kk} \end{bmatrix}$$

$$= -\cos \theta a_{ii} \sin \theta - \sin^2 \theta a_{ki} + \cos \theta \sin \theta a_{ik} + \cos \theta \sin \theta a_{kk} = 0$$

$$\cos \theta \sin \theta [a_{kk} - a_{ii}] + a_{ik} [\sin 2\theta - \sin^2 \theta] = 0$$

$$\frac{1}{2} \sin 2\theta [a_{kk} - a_{ii}] + a_{ik} [\cos 2\theta] = 0$$

$$a_{ik} \cos 2\theta = -\frac{1}{2} \sin 2\theta [a_{kk} - a_{ii}]$$

$$\frac{2a_{ik}}{a_{ii} - a_{kk}} = \frac{\sin 2\theta}{\cos 2\theta}$$

$$\frac{2a_{jk}}{a_{ii} - a_{kk}} = \tan \theta$$

$$\theta = \frac{1}{2} \tan^{-1} \left[\frac{2a_{jk}}{a_{ii} - a_{kk}} \right]$$

Find Eigen Value and Eigen vector for the symmetric matrix

$$A = \begin{bmatrix} 1 & \sqrt{2} & 2 \\ \sqrt{2} & 3 & \sqrt{2} \\ 2 & \sqrt{2} & 1 \end{bmatrix}$$

Sol: The largest non diagonal element is $a_{13} = 2 = a_{31}$
 $a_{11} = 1$ $a_{33} = 1$

$$\theta = \frac{1}{2} \tan^{-1} \left[\frac{2a_{jk}}{a_{ii} - a_{kk}} \right] = \frac{1}{2} \tan^{-1} \left[\frac{2(2)}{1-1} \right] = \frac{1}{2} \times \frac{\pi}{2} = \underline{\underline{\pi/4}}$$

$$S = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}$$

$$B = S^{-1} A S = \begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}^{-1} \begin{bmatrix} 1 & \sqrt{2} & 2 \\ \sqrt{2} & 3 & \sqrt{2} \\ 2 & \sqrt{2} & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 1 \\ 2 & 3 & 0 \\ 4.5 & 3 & -0.5 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 0 \\ 2 & 3 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

The largest & non diagonal element is $a_{12}=2, a_{11}=3, a_{22}=3$

$$\theta = \frac{1}{2} \tan^{-1} \left[\frac{2a_{12}}{a_{11} - a_{22}} \right] = \frac{1}{2} \tan^{-1} \left(\frac{2(2)}{3-3} \right) = \pi/4$$

$$S_2 = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$B_1 = S_2^{-1} B S_2$$

$$= \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 3 & 2 & 0 \\ 2 & 3 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 2.5 & 0 \\ 0 & 1.5 & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Eigen vectors

$$S = S_1 S_2 = \begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1/2 & -1/2 & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/2 & -1/2 & 1/\sqrt{2} \end{bmatrix}$$

Given's method:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad S = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix} \quad \tan\theta = \frac{a_{13}}{a_{12}}$$

$B = S^{-1}AS$

$$B = \begin{bmatrix} a_1 & b_2 & 0 \\ b_2 & a_2 & b_3 \\ 0 & b_3 & a_3 \end{bmatrix}$$

$$|B - \lambda I| = 0$$

$$\begin{vmatrix} a_1 - \lambda & b_2 & 0 \\ b_2 & a_2 - \lambda & b_3 \\ 0 & b_3 & a_3 - \lambda \end{vmatrix} = 0$$

$$P_0(\lambda) = 1$$

$$P_1(\lambda) = a_1 - \lambda$$

$$P_{k+1}(\lambda) = (a_{k+1} - \lambda)P_k(\lambda) - b_{k+1}^2 P_{k-1}(\lambda)$$

$$k=1, 2, 3, \dots$$

$$(B - \lambda I)X = 0$$

$$\begin{vmatrix} a_1 - \lambda & b_2 & 0 \\ b_2 & a_2 - \lambda & b_3 \\ 0 & b_3 & a_3 - \lambda \end{vmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$(a_1 - \lambda)x_1 + b_2x_2 + 0x_3 = 0$$

$$b_2x_1 + (a_2 - \lambda)x_2 + b_3x_3 = 0$$

$$0x_1 + b_3x_2 + (a_3 - \lambda)x_3 = 0$$

$$a_1x_1 + b_2x_2 + 0x_3 = 0$$

$$b_2x_1 + a_2x_2 + b_3x_3 = 0$$

$$\frac{b_2x_1}{b_2} = \frac{-x_2}{1} = \frac{x_3}{\frac{b_3}{b_2}}$$

4x4

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

$$S = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & 0 & \sin\theta \\ 0 & 0 & 1 & 0 \\ 0 & \sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$\tan\theta = \frac{a_{14}}{a_{13}}$$

$$B = S^{-1}AS$$

$$= \begin{bmatrix} a_{11} & a_{12} & a_{13} & 0 \\ a_{21} & a_{22} & a_{23} & 0 \\ 0 & 0 & a_{33} & 0 \\ 0 & 0 & a_{43} & a_{44} \end{bmatrix}$$

1) Find the eigen value of the symmetric matrix $A = \begin{bmatrix} 2 & 3 & -1 \\ 3 & 1 & 2 \\ -1 & 2 & -1 \end{bmatrix}$ using Gram's method

$$\tan \theta = \frac{a_{13}}{a_{12}} = \frac{-1}{3} = -0.3218$$

$$S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.9487 & 0.3163 \\ 0 & -0.3163 & 0.9487 \end{bmatrix}$$

$$B = S_1^{-1}AS_1$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.9487 & 0.3163 \\ 0 & -0.3163 & 0.9487 \end{bmatrix}^{-1} \begin{bmatrix} 2 & 3 & -1 \\ 3 & 1 & 2 \\ -1 & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.9487 & 0.3163 \\ 0 & -0.3163 & 0.9487 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 3.1624 & 0.0002 \\ 3.1624 & -0.4 & 2.1999 \\ 0 & 2.2 & 0.4002 \end{bmatrix}$$

The Sturm sequence is given by $|B - \lambda I| = 0$

$$\begin{vmatrix} 2-\lambda & 3.1624 & 0 \\ 3.1624 & -0.4-\lambda & 2.2 \\ 0 & 2.2 & 0.4-\lambda \end{vmatrix} = 0$$

$$P_0(\lambda) = 1$$

$$P_1(\lambda) = a_1 - \lambda = 2 - \lambda$$

$$P_{k+1}(\lambda) = (a_{k+1} - \lambda) P_k(\lambda) - b_{k+1}^2 P_{k-1}(\lambda)$$

$$P_2(\lambda) = (a_2 - \lambda) P_1(\lambda) - b_2^2 P_0(\lambda)$$

$$= (-0.4 - \lambda)(2 - \lambda) - (3.1623)^2 (1)$$

$$= -0.4$$

$$= -0.8 + 0.4\lambda - 2\lambda + \lambda^2 = 10.0001$$

$$= \lambda^2 - 1.6\lambda - 10.8$$

$$P_3(\lambda) = (a_3 - \lambda) P_2(\lambda) - b_3^2 P_1(\lambda)$$

$$= (0.4 - \lambda)(\lambda^2 - 1.6\lambda - 10.8) - (2.2)^2 (2 - \lambda)$$

$$= 0.4\lambda^2 - 0.64\lambda - 4.3200 - \lambda^3 + 1.6\lambda^2 + 10.8\lambda - (4.84)(2 - \lambda)$$

$$= -\lambda^3 + 2\lambda^2 + 15\lambda - 14 = 0$$

$$= -\lambda^3 + 2\lambda^2 + 15\lambda - 14 = 0$$

$$= \lambda^3 - 2\lambda^2 + \begin{pmatrix} 0.4 \\ -0.4 & 2.2 \\ 2.2 & 0.4 \end{pmatrix} + \begin{pmatrix} 2 & 10 \\ 0 & 0.4 \end{pmatrix} + \begin{pmatrix} 2 & 3.1623 \\ 3.1623 & -0.4 \end{pmatrix} \lambda - |A| = 0$$

$$\therefore -\lambda^3 + 2\lambda^2 + 15\lambda - 14 = 0$$

$$\lambda_1 = -3.4753, \lambda_2 = 4.5995, \lambda_3 = 0.8758$$

2) Transfer the matrix $A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 1 \end{bmatrix}$ to the diagonal form by given method & hence find eigen value of A .

$$\tan \theta = \frac{a_{13}}{a_{12}} = \frac{4}{2} = 2 \Rightarrow \theta = 1.1071$$

$$S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.4473 & -0.8944 \\ 0 & +0.8944 & 0.4473 \end{bmatrix}$$

$$B = S_1^{-1} A S_1$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.4473 & -0.8944 \\ 0 & +0.8944 & 0.4473 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.4473 & -0.8944 \\ 0 & -0.8944 & 0.4473 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4.4722 & 0 \\ 4.4721 & 2.6 & -1.1997 \\ 0 & -1.2 & -0.6 \end{bmatrix}$$

Sturm sequence $|B - \lambda I| = 0$

$$= \begin{vmatrix} 1-\lambda & 4.4722 & 0 \\ 4.4721 & 2.6-\lambda & -1.2 \\ 0 & -1.2 & -0.6-\lambda \end{vmatrix} = 0$$

$$P_0(\lambda) = 1$$

$$P_1(\lambda) = a_1 - \lambda = 1 - \lambda$$

$$P_{k+1}(\lambda) = (a_{k+1} - \lambda) P_k(\lambda) - b_{k+1}^2 P_{k-1}(\lambda)$$

$$P_2(\lambda) = (a_2 - \lambda) P_1(\lambda) - b_2^2 P_0(\lambda)$$

$$= (2.6 - \lambda)(1 - \lambda) - (4.4721)^2 (1)$$

$$= 2.6 - 2.6\lambda - \lambda + \lambda^2 - 19.9997$$

$$= \lambda^2 - 3.6\lambda - 17.3997$$

$$P_3(\lambda) = (a_3 - \lambda) P_2(\lambda) - b_3^2 (P_1(\lambda))$$

$$= (-0.6 - \lambda)(\lambda^2 - 3.6\lambda - 17.3997) - (-1.2)^2 (1 - \lambda)$$

$$= -0.6\lambda^2 + 2.16\lambda + 10.4398 - \lambda^3 + 3.6\lambda^2 + 17.3997\lambda - 1.44 + 1.44\lambda$$

$$= -\lambda^3 + 3\lambda^2 + 20.9997\lambda + 8.9998$$

$$= \lambda^3 - 3\lambda^2 - 21\lambda - 9$$

$$\lambda_1 = 6.46, \lambda_2 = -0.4641, \lambda_3 = -3$$

Rutisha hammer method for arbitrary matrix.

$$A = A_1, A_1 = L_1 U_1, l_{ii} = 1$$

$$L_1 = \begin{bmatrix} 1 & 0 \\ l_{21} & 1 \end{bmatrix}, U = \begin{bmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{bmatrix}$$

$$A_1 = L_1 U_1$$

$$= \begin{bmatrix} 1 & 0 \\ l_{21} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{bmatrix}$$

$$= \begin{bmatrix} u_{11} & u_{12} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} \end{bmatrix}$$

$$A_2 = U_1 L_1$$

$$A_2 = L_2 U_2$$

$$A_3 = U_2 L_2$$

$$A_3 = L_3 U_3$$

$$A_4 = U_3 L_3$$

$$A_4 = L_4 U_4$$

Find all the eigen values of the matrix: $A = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$ using
Perron's lower method.

Sol:- let $A_1 = L_1 U_1$

$$\begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \lambda_{21} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{bmatrix}$$

$$\begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} \\ \lambda_{21} u_{11} & \lambda_{21} u_{12} + u_{22} \end{bmatrix}$$

$$u_{11} = 4, \quad u_{12} = 2, \quad \lambda_{21} u_{11} = -1$$

$$\lambda_{21}(4) = -1$$

$$\lambda_{21} = -1/4$$

$$\lambda_{21} u_{12} + u_{22} = 1$$

$$\left(-\frac{1}{4}\right)(2) + u_{22} = 1$$

$$-1/2 + u_{22} = 1 + 1/2$$

$$u_{22} = 3/2$$

$$U_1 = \begin{bmatrix} 4 & 2 \\ -1/4 & 3/2 \end{bmatrix}$$

$$A_2 = U_1 L_1$$

$$= \begin{bmatrix} 4 & 2 \\ 0 & 3/2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1/4 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 4 - 1/2 & 2 \\ -3/8 & 3/2 \end{bmatrix} = \begin{bmatrix} 7/2 & 2 \\ -3/8 & 3/2 \end{bmatrix}$$

$$A_2 = L_2 U_2$$

$$\begin{bmatrix} 7/2 & 2 \\ -3/8 & 3/2 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} \\ \lambda_{21} u_{11} & \lambda_{21} u_{12} + u_{22} \end{bmatrix}$$

$$u_{11} = 7/2, \quad u_{12} = 2,$$

$$\lambda_{21} u_{11} = -3/8$$

$$\lambda_{21} (7/2) = -3/8$$

$$\lambda_{21} = -\frac{6}{56} = -3/28$$

$$\lambda_{21} u_{12} + u_{22} = 3/2$$

$$\left(-\frac{3}{28}\right)2 + u_{22} = 3/2$$

$$u_{22} = 12/7$$

$$A_3 = U_2 L_2$$

$$= \begin{bmatrix} 7/2 & 2 \\ 0 & 12/7 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -3/28 & 1 \end{bmatrix} = \begin{bmatrix} 23/7 & 2 \\ -9/49 & 12/7 \end{bmatrix}$$

$$A_3 = L_3 U_3$$

$$= \begin{bmatrix} 23/7 & 2 \\ -9/49 & 12/7 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} \end{bmatrix}$$

$$u_{11} = 23/7, \quad u_{12} = 2, \quad l_{21} = -9/161$$

$$u_{22} = 12/7 - l_{21}u_{12}$$

$$= 12/7 - \left(-\frac{9}{161}\right)(2)$$

$$= 4\frac{2}{23}$$

$$A_4 = U_3 L_3$$

$$= \begin{bmatrix} 23/7 & 2 \\ -9/49 & 12/7 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -9/161 & 1 \end{bmatrix} = \begin{bmatrix} 3.1739 & 2 \\ -0.1021 & 1.8241 \end{bmatrix}$$

$$A_4 = L_4 U_4$$

$$\begin{bmatrix} 3.1739 & 2 \\ -0.1021 & 1.8241 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} \end{bmatrix}$$

$$u_{11} = 3.1739, \quad u_{12} = 2, \quad l_{21} = \frac{-0.1021}{3.1739}$$

$$= -0.0322 \approx 0$$

$$u_{22} = 1.8241 - (-0.0322)(2)$$

$$= 1.8885$$

$$A_5 = U_4 L_4$$

$$= \begin{bmatrix} 3.1739 & 2 \\ -0.0322 & 1.8885 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0.0322 & 1 \end{bmatrix} = \begin{bmatrix} 3.1095 & 2 \\ -0.0608 & 1.8885 \end{bmatrix}$$

$$A = LU$$

$$\begin{bmatrix} 3.1095 & 2 \\ -0.0608 & 1.8885 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} \end{bmatrix}$$

$$u_{11} = 3.1095, u_{12} = 2, l_{21} = \frac{-0.0608}{3.1095}$$

$$= \underline{\underline{-0.1955}}$$

$$l_{21}u_{12} + u_{22} = 1.8885$$

$$u_{22} = 1.8885 - (-0.1955)(2)$$

$$= \underline{\underline{2.2795}}$$

$$A = U L$$

$$= \begin{bmatrix} 3.1095 & 2 \\ 0 & 2.2795 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -0.1955 & 1 \end{bmatrix}$$

Find all the eigen values of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \\ 1 & 3 & 2 \end{bmatrix}$ using Rutisha's method.
 iterate till the lower diagonal parts are less than 0.05 in magnitude.

Sol: Let $A_1 = L_1 U_1$

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \\ 1 & 3 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ u_{11}l_{21} & u_{12}l_{21} + u_{22} & u_{13}l_{21} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

$$u_{11} = 1, \quad u_{12} = 1, \quad u_{13} = 1, \quad u_{11}l_{21} = 2, \quad l_{21}u_{12} + u_{22} = 1, \quad u_{22} = -1$$

$$l_{21} = 2$$

$$l_{31}u_{11} = 1, \quad l_{31}u_{12} + l_{32}u_{22} = 3, \quad l_{31}u_{13} + l_{32}u_{23} + u_{33} = 2$$

$$l_{31} = 1, \quad 1 + (-1)l_{32} = 3, \quad (1)(1) + (-2)(-1) + u_{33} = 2$$

$$l_{32} = -2, \quad u_{33} = 1$$

$$u_{13}l_{21} + u_{23} = 2$$

$$(1)(2) + u_{23} = 2$$

$$u_{23} = 0$$

$$A_2 = U_1 L_1$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -1 & 1 \\ -2 & -1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$$

$$A_2 = L_2 U_2$$

$$\begin{bmatrix} 4 & -1 & 1 \\ -2 & -1 & 0 \\ 1 & -2 & 1 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ u_{11}l_{21} & u_{12}l_{21} + u_{22} & u_{13}l_{21} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

$$u_{11} = 4, \quad u_{12} = -1, \quad u_{13} = 1$$

$$\lambda_{21} u_{11} = -2$$

$$\lambda_{21} u_{12} + u_{22} = -1$$

$$\lambda_{21} = -\frac{1}{2}$$

$$(-\frac{1}{2})(-1) + u_{22} = -1$$

$$\lambda_{21} u_{13} + u_{23} = 0$$

$$u_{22} = -1 - \frac{1}{2}$$

$$u_{22} = -\frac{3}{2}$$

$$(-\frac{1}{2})(1) + u_{23} = 0$$

$$\lambda_{31} u_{11} = 1$$

$$\lambda_{31} u_{11} + \lambda_{32} u_{22} = -2$$

$$u_{23} = \frac{1}{2}$$

$$\lambda_{31} = \frac{1}{4}$$

$$(\frac{1}{4})(-1) + \lambda_{32}(-\frac{3}{2}) = -2$$

$$\lambda_{31} u_{13} + \lambda_{32} u_{23} + u_{33} = 1$$

$$\lambda_{32} = \frac{-2 + \frac{1}{4}}{-\frac{3}{2}}$$

$$(\frac{1}{4})(1) + (\frac{7}{6})(\frac{1}{2}) + u_{33} = 1$$

$$\lambda_{32} = \frac{-7}{\cancel{6} \times \frac{1}{\cancel{3}}} = \frac{7}{6}$$

$$u_{33} = 1 - \frac{1}{4} - \frac{7}{6} = \frac{1}{6}$$

$$= \frac{3}{4} \times \frac{6}{7}$$

$$A_3 = U_2 L_2$$

$$= \begin{bmatrix} 4 & -1 & 1 \\ 0 & -\frac{3}{2} & \frac{1}{2} \\ 0 & 0 & \frac{1}{6} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ \frac{1}{4} & \frac{7}{6} & 1 \end{bmatrix} = \begin{bmatrix} \frac{19}{4} & \frac{1}{6} & 1 \\ \frac{7}{8} & -\frac{1}{12} & \frac{1}{2} \\ \frac{1}{24} & \frac{7}{36} & \frac{1}{6} \end{bmatrix}$$

$$A_3 = L_3 U_3$$

$$= \begin{bmatrix} \frac{19}{4} & \frac{1}{6} & 1 \\ \frac{7}{8} & -\frac{1}{12} & \frac{1}{2} \\ \frac{1}{24} & \frac{7}{36} & \frac{1}{6} \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ \lambda_{21} u_{11} & \lambda_{21} u_{12} + u_{22} & \lambda_{21} u_{13} + u_{23} \\ \lambda_{31} u_{11} & \lambda_{31} u_{12} + \lambda_{32} u_{22} & \lambda_{31} u_{13} + \lambda_{32} u_{23} + u_{33} \end{bmatrix}$$

$$u_{11} = 19/4 \quad u_{12} = 1/6 \quad u_{13} = 1 \quad l_{21} u_{11} = 7/8 \quad l_{21} u_{12} + u_{22} = 11/12$$

$$l_{21} = 7/8 \times 4/19 \quad (7/38)(1/6) + u_{22} = 11/12$$

$$= 7/38$$

$$l_{21} u_{13} + u_{23} = 1/2 \quad l_{31} u_{11} = 1/24 \quad u_{22} = 11/12 - 7/228$$

$$7/38(1) + u_{23} = 1/2 \quad l_{31} = 1/24 \times 4/19 = 1/114$$

$$u_{23} = 6/19$$

$$l_{31} u_{12} + l_{32} u_{22} = 7/36 \quad l_{31} u_{13} + l_{32} u_{23} + u_{33} = 1/2$$

$$(1/114)(1/6) + (-18/19)l_{32} = 7/36 \quad u_{33} =$$

$$l_{32} = -11/54 \quad (-11/54)(1) + (-11/54)(6/19) + u_{33} = 1/2$$

$$u_{33} = 2/9$$

$$A_4 = U_3 L_3$$

$$= \begin{bmatrix} 19/4 & 1/6 & 1 \\ 7/8 & -11/9 & 6/19 \\ 0 & 0 & 2/9 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 7/38 & 1 & 0 \\ 1/114 & -11/54 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 4.7895 & -0.03704 & 1 \\ -0.1717 & -1.0117 & 0.3158 \\ 0.0019 & -0.0452 & 0.2222 \end{bmatrix}$$

$$A_4 = L_4 U_4$$

$$\begin{bmatrix} 4.7895 & -0.03708 & 1 \\ -0.1717 & -1.0117 & 0.3158 \\ 0.0019 & -0.0452 & 0.2222 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12}+u_{22} & l_{21}u_{13}+u_{23} \\ l_{31}u_{11} & l_{31}u_{12}+l_{32}u_{22} & l_{31}u_{13}+l_{32}u_{23}+u_{33} \end{bmatrix}$$

$$u_{11} = 4.7895, \quad u_{12} = -0.03708, \quad u_{13} = 1$$

$$l_{21}u_{11} = -0.1717, \quad l_{21}u_{12}+u_{22} = -1.0117$$

$$l_{21} = -0.0358, \quad (-0.0358)(-0.03708)+u_{22} = -1.0117$$

$$u_{22} = \underline{\underline{-1.0123}}$$

$$l_{31}u_{13}+u_{23} = 0.3158$$

$$l_{31}u_{11} = 0.0019$$

$$(-0.0358)(1)+u_{23} = 0.3158$$

$$l_{31} = \underline{\underline{0.0004}}$$

$$u_{23} = \underline{\underline{-8.8242 \times 10^{-5}}}$$

$$l_{31}u_{12}+l_{32}u_{22} = -0.0452$$

$$l_{31}u_{13}+l_{32}u_{23}+u_{33} = 0.2222$$

$$(0.0004)(-0.03708)+l_{32}(-1.0123) = -0.0452$$

$$u_{33} = \underline{\underline{0.206}}$$

$$l_{32} = \underline{\underline{0.0445}}$$

$$A_5 = U_4 L_4$$

$$\begin{bmatrix} 4.7895 & -0.037 & 1 \\ 0 & -1.0123 & 0.3154 \\ 0 & 0 & 0.206 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -0.0358 & 1 & 0 \\ 0.0004 & 0.0445 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 4.7912 & 0.0075 & 1 \\ 0.0365 & -0.997 & 0.3514 \\ 0.0001 & 0.011 & 0.206 \end{bmatrix}$$

$$A = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$$

Householder's method:

$$A = \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}_{4 \times 4}$$

Ist rotation

$$A_2 = S_1 A_1 S_1$$

$$S_1 = I - 2V_1 V_1^T$$

$$V_1 = [0 \ S_2 \ S_3 \ S_4]^T$$

$$S_2^2 = \frac{1}{2} \left[1 - \frac{a_{12} \times \text{sign}(a_{12})}{q} \right]$$

$$S_3 = \frac{a_{13} \times \text{sign}(a_{12})}{2q \times S_2}$$

$$S_4 = \frac{a_{14} \times \text{sign}(a_{12})}{2q \times S_2}$$

$$q = \sqrt{a_{12}^2 + a_{13}^2 + a_{14}^2}$$

IInd rotation

$$A_3 = S_2 A_2 S_2$$

$$S_2 = I - 2V_2 V_2^T$$

$$V_2 = [0 \ 0 \ S_3 \ S_4]^T$$

$$S_3^2 = \frac{1}{2} \left[1 - \frac{a_{23} \times \text{sign}(a_{23})}{q} \right]$$

$$S_4 = \frac{a_{24} \times \text{sign}(a_{23})}{2q \times S_3}$$

$$q = \sqrt{a_{23}^2 + a_{24}^2}$$

For 3x3 matrix

$$A_1 = \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}_{3 \times 3}$$

Ist rotation

$$A_2 = S_1 A_1 S_1$$

$$S_1 = I - 2V_1 V_1^T$$

$$V_1 = [0 \ S_2 \ S_3]^T$$

$$S_2^2 = \frac{1}{2} \left[1 - \frac{a_{12} \times \text{sign}(a_{12})}{q} \right]$$

$$S_3 = \frac{a_{13} \times \text{sign}(a_{12})}{2q \times S_2}$$

$$q = \sqrt{a_{12}^2 + a_{13}^2}$$

1) Use the householder method to reduce the matrix.

$$A_1 = \begin{bmatrix} 2 & -1 & -1 & 1 \\ -1 & 4 & 1 & -1 \\ -1 & 1 & 3 & -1 \\ 1 & -1 & -1 & 2 \end{bmatrix} \text{ into the tridiagonal form.}$$

Sol. T^H rotation.

$$A_2 = S_1 A_1 S_1$$

$$S_1 = I - 2V_1 V_1^T$$

$$V_1 = [0, S_2, S_3, S_4]^T$$

$$S_2 = \frac{1}{2} \left[1 + \frac{a_{12} \times \text{sign}(a_{12})}{q} \right]$$

$$= \frac{1}{2} \left[1 + \frac{(-1)(-)}{\sqrt{3}} \right]$$

$$S_2 = 0.8881$$

$$S_3 = \frac{a_{13} \times \text{sign}(a_{12})}{2q \times S_2}$$

$$= \frac{(-1)(-)}{2\sqrt{3} \times 0.8881} = 0.3250$$

$$S_4 = \frac{a_{14} \times \text{sign}(a_{12})}{2q \times S_2} = \frac{1 \times (-1)}{2\sqrt{3} \times 0.8881} = -0.3250$$

$$V_1 = [0.08881, 0.3250, -0.3250]^T$$

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$$S_1 = I - 2V_1 V_1^T$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} - 2 \begin{bmatrix} 0 & 0.8881 & 0.3250 & -0.3250 \end{bmatrix} \begin{bmatrix} 0.8881 \\ 0.3250 \\ -0.3250 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} - 2 \begin{bmatrix} 0 \\ 0.8881 \\ 0.3250 \\ -0.3250 \end{bmatrix} \begin{bmatrix} 0 & 0.8881 & 0.3250 & -0.3250 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 1.7762 \\ 0.65 \\ -0.65 \end{bmatrix} \begin{bmatrix} 0 & 0.8881 & 0.3250 & -0.3250 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1.5774 & 0.5773 & -0.5773 \\ 0 & 0.5773 & 0.2113 & -0.2113 \\ 0 & -0.5773 & -0.2113 & 0.2113 \end{bmatrix}$$

$$S_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -0.5774 & -0.5773 & 0.5773 \\ 0 & -0.5773 & 0.7887 & 0.2113 \\ 0 & 0.5773 & 0.2113 & 0.7887 \end{bmatrix}$$

$$A_2 = S_1 A_1 S_1$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -0.5774 & -0.5774 & 0.5774 \\ 0 & -0.5774 & 0.7887 & 0.2113 \\ 0 & 0.5774 & 0.2113 & 0.7887 \end{bmatrix} \begin{bmatrix} 2 & -1 & -1 & 1 \\ -1 & 4 & 1 & -1 \\ -1 & 1 & 3 & -1 \\ 1 & -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -0.5774 & -0.5774 & 0.5774 \\ 0 & -0.5774 & 0.7887 & 0.2113 \\ 0 & 0.5774 & 0.2113 & 0.7887 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -1 & -1 & 1 \\ 1.7372 & -3.4644 & -2.8870 & 2.3096 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$= 2 \quad 1.732 \quad 0 \quad 0$$

$$1.732 \quad 5 \quad 0.2113 \quad -0.7787$$

$$0 \quad 0.2113 \quad 2.8865 \quad -0.5$$

$$0 \quad -0.7787 \quad -0.5 \quad 1.7113$$

2nd rotation:

$$A_3 = S_2 A_1 S_2$$

$$S_2 = I - 2V_2 V_2^T$$

$$V_2 = [0 \quad 0 \quad S_3 \quad S_4]^T$$

$$S_3^2 = \frac{1}{2} \left[1 + \frac{a_{23} \text{sign}(a_{23})}{q} \right]$$

$$= \frac{1}{2} \left[1 + \frac{0.2113}{0.8069} \right]$$

$$= \underline{\underline{\frac{1}{2} 0.7943}}$$

$$q = \sqrt{a_{23}^2 + a_{24}^2}$$

$$= \sqrt{0.2113^2 + (-0.7787)^2}$$

$$= \underline{\underline{0.8069}}$$

$$S_4 = \frac{a_{24} \times \text{sign}(a_{23})}{2q \times S_3}$$

$$S_4 =$$

$$= \frac{-0.7787}{2 \times 0.8069 \times 0.7943}$$

$$= -0.6076$$

$$S_4 = \underline{\underline{-0.6076}}$$

$$V_2 = [0 \ 0 \ s_3 \ s_4]^T$$

$$= [0 \ 0 \ 0.7943 \ -0.6076]^T$$

$$S_1 = I - 2V_2V_2^T$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} - 2 \begin{bmatrix} 0 \\ 0 \\ 0.7943 \\ -0.6076 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0.7943 & -0.6076 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} - 2 \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0.6309 & -0.4826 \\ 0 & 0 & 0.5886 & -0.4826 \\ 0 & 0 & -0.4826 & 0.3692 \end{bmatrix}$$

$$S_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -0.2618 & 0.9652 \\ 0 & 0 & 0.9652 & 0.2616 \end{bmatrix}$$

$$A_3 = S_2 A S_2$$

$$= \begin{bmatrix} 2 & 1.732 & 0 & 0 \\ 1.732 & 5 & -0.816 & 0.0002 \\ 0 & -0.816 & 2.0448 & -0.728 \\ 0 & -0.0024 & -0.728 & 2.5538 \end{bmatrix}$$

2) Using Householder method reduce matrix A into
 $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ into a bidiagonal form & hence find eigen values

1st rotation

$$A = S_1 A S_1^T$$

$$S_1 = I - 2 V_1 V_1^T$$

$$V_1 = [0 \ S_2 \ S_3]^T$$

$$S_2 = \frac{1}{2} \left[1 + \frac{a_{12} \times \text{sign}(a_{12})}{q} \right]$$

$$= \frac{1}{2} \left[1 + \frac{1 \times (+)}{\sqrt{2}} \right]$$

$$S_2 = \underline{0.9239}$$

$$S_3 = \frac{a_{13} \times \text{sign}(a_{13})}{2 \times q \times S_2}$$

$$= \frac{1 \times (+)}{2 \times \sqrt{2} \times 0.9239}$$

$$= \underline{0.3827}$$

$$V_1 = [0 \ 0.9239 \ 0.3827]^T$$

$$S_1 = I - 2 V_1 V_1^T$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - 2 \begin{bmatrix} 0 \\ 0.9239 \\ 0.3827 \end{bmatrix} \begin{bmatrix} 0 & 0.9239 & 0.3827 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - 2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0.8536 & 0.3536 \\ 0 & 0.3536 & 0.1465 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & -0.7072 & -0.7072 \\ 0 & -0.7072 & -0.2930 \end{bmatrix}$$

$$A_2 = S_1 A S_1$$

$$P_1 = \begin{bmatrix} 2 & -1.4144 & -1.4144 \\ 1.0002 & 1.0003 & \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -1.4144 & 0 \\ -1.4144 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The Sturm sequence is given by $|B - \lambda I| = 0$

$$\begin{vmatrix} 2-\lambda & -1.4144 & 0 \\ -1.4144 & 1-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{vmatrix} = 0$$

$$P_0(\lambda) = 1$$

$$P_1(\lambda) = a_1 - \lambda = 2 - \lambda$$

$$P_{k+1}(\lambda) = (a_{k+1} - \lambda) P_k(\lambda) - b_{k+1}^2 P_{k-1}(\lambda)$$

$$P_2(\lambda) = (1 - \lambda)(2 - \lambda) - (-1.4144)^2 (1)$$

$$= 2 - \lambda - 2\lambda + \lambda^2 - 2.0005$$

$$= \lambda^2 - 3\lambda - 2.0005 + 0$$

$$\begin{aligned}
 P_3(\lambda) &= (a_3 - \lambda) P_2(\lambda) - b_3^2 P_1(\lambda) \\
 &= (1 - \lambda)(\lambda^2 - 3\lambda) - 0 \\
 &= \lambda^2 - 3\lambda - \lambda^3 + 3\lambda^2 \\
 &= -\lambda^3 + 4\lambda^2 - 3\lambda = 0
 \end{aligned}$$

$$\underline{\underline{\lambda = 3, 1, 0}}$$

$$3) A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & -1 \\ 3 & -1 & 1 \end{bmatrix}$$

$$\begin{aligned}
 q &= \sqrt{a_{12}^2 + a_{13}^2} \\
 &= \sqrt{2^2 + 3^2} = \sqrt{13}
 \end{aligned}$$

$$S_2 = \frac{1}{2} \left[1 + \frac{a_{12} \times \text{sign}(a_{12})}{q} \right]$$

$$= \frac{1}{2} \left[1 + \frac{2}{\sqrt{13}} \right] = 0.8817$$

$$S_3 = \frac{a_{13} \times \text{sign}(a_{12})}{2 \times q \times S_2}$$

$$= \frac{3}{2 \times \sqrt{13} \times 0.8817} = 0.4718$$

$$V_1 = \begin{bmatrix} 0 & S_2 & S_3 \end{bmatrix}^T$$

$$V_1 = \begin{bmatrix} 0 & 0.8817 & 0.4718 \end{bmatrix}^T$$

$$S_1 = I - 2V_1 V_1^T$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - 2 \begin{bmatrix} 0 \\ -0.8817 \\ 0.4718 \end{bmatrix} \begin{bmatrix} 0 & 0.8817 & 0.4718 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 1.7634 \\ 0.9436 \end{bmatrix} \begin{bmatrix} 0 & 0.8817 & 0.4718 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1.5548 & 0.8320 \\ 0 & 0.8320 & 0.4452 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & -0.5548 & -0.8320 \\ 0 & -0.8320 & 0.5548 \end{bmatrix}$$

$$A_2 = S_1 A S_1$$

$$= \begin{bmatrix} 1 & -3.6056 & 0.0004 \\ -3.6056 & 0.0768 & -0.3844 \\ 0.0004 & -0.3844 & 1.9232 \end{bmatrix}$$

The Sturm sequence

$$\begin{vmatrix} 1-\lambda & -3.6056 & 0.0004 \\ -3.6056 & 0.0768-\lambda & -0.3844 \\ 0.0004 & -0.3844 & 1.9232-\lambda \end{vmatrix} = 0$$

$$P_0(\lambda) = 1$$

$$P_1(\lambda) = a_1 - \lambda = 1 - \lambda$$

$$P_{k+1}(\lambda) = (a_{k+1} - \lambda) P_k(\lambda) - b_{k+1}^2 P_{k-1}(\lambda)$$

$$= (0.0768 - \lambda)(1 - \lambda) - (3.6056)^2 (1)$$

$$= 0.0768 - 0.0768\lambda - \lambda + \lambda^2 - 13.0004$$

$$= \lambda^2 - 1.0768\lambda - 13 = 0$$

$$P_3(\lambda) = (a_3 - \lambda)P_2(\lambda) - b_3^2 P_1(\lambda)$$

$$= (0.9232 - \lambda)(\lambda^2 - 1.0768\lambda - 13) - (-0.3844)^2(1 - \lambda)$$

$$= 0.9232\lambda^2 - 2.0709\lambda - 25.0016 - \lambda^3 + 1.0768\lambda^2 + 13\lambda - 0.1478(1 - \lambda)$$

$$= -\lambda^3 + 3\lambda^2 + 11.1478\lambda - 25$$

$$= -\lambda^3 + 3\lambda^2 + 11.1478\lambda - 25.1478$$

$$\lambda_1 = -3.12, 4.2299, 1.8997$$